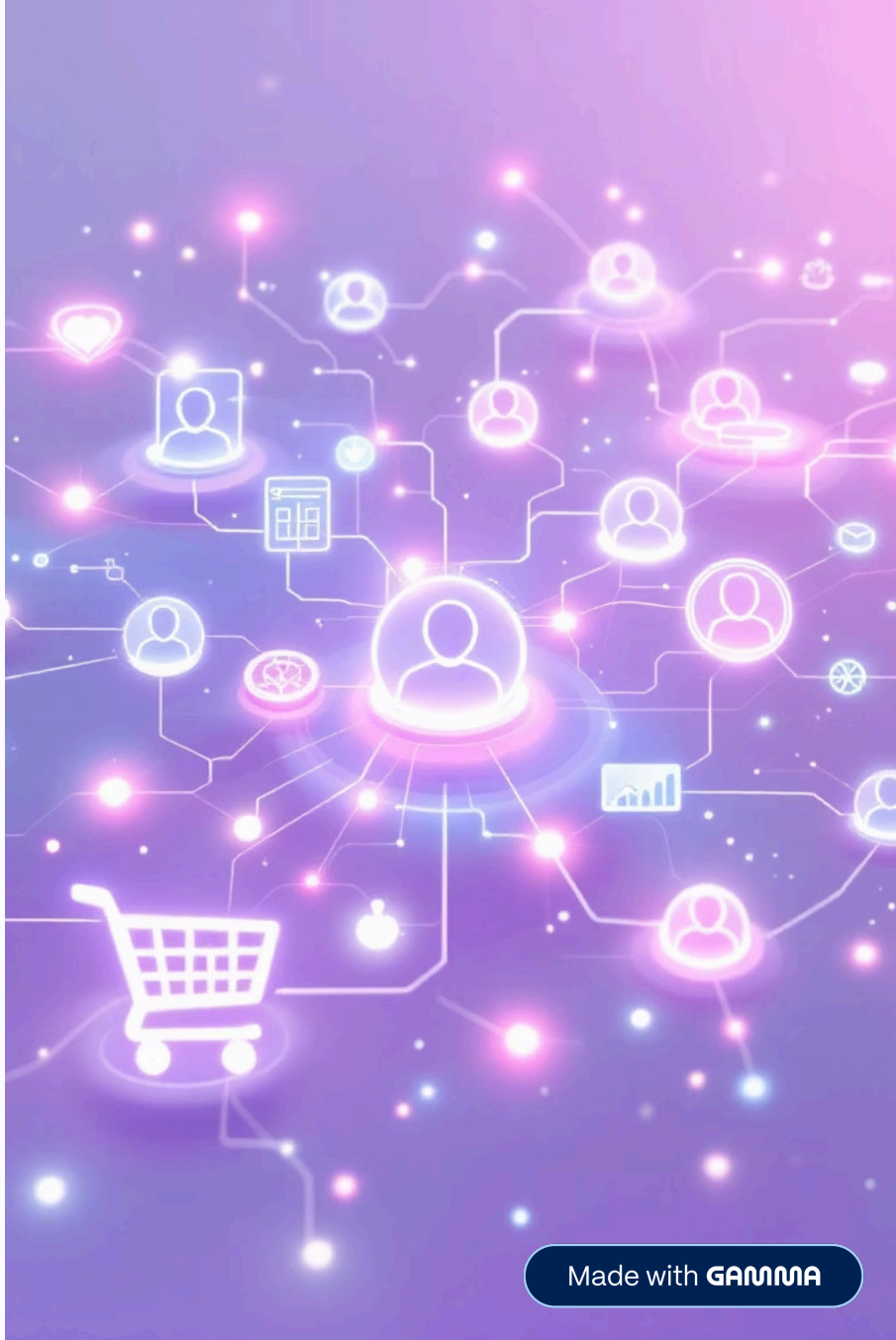
Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



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Project Overview

1

Data Analysis

3,900 purchases across categories.

2

Key Insights

Spending patterns, segments, product preferences, subscriptions.

3

Strategic Goals

Optimize operations, reduce costs, improve forecasting.

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Dataset Summary

Data at a Glance

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in Review Rating

Key Features

- Customer demographics (Age, Gender, Location, Subscription)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)

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Exploratory Data Analysis (Python)

01

Data Preparation

Loading, initial exploration, missing data handling (median imputation).

02

Column Standardization

Renamed columns to snake case.

03

Feature Engineering

Created age_group and purchase_frequency_days.

04

Data Consistency

Dropped redundant 'promo_code_used'.

05

Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.

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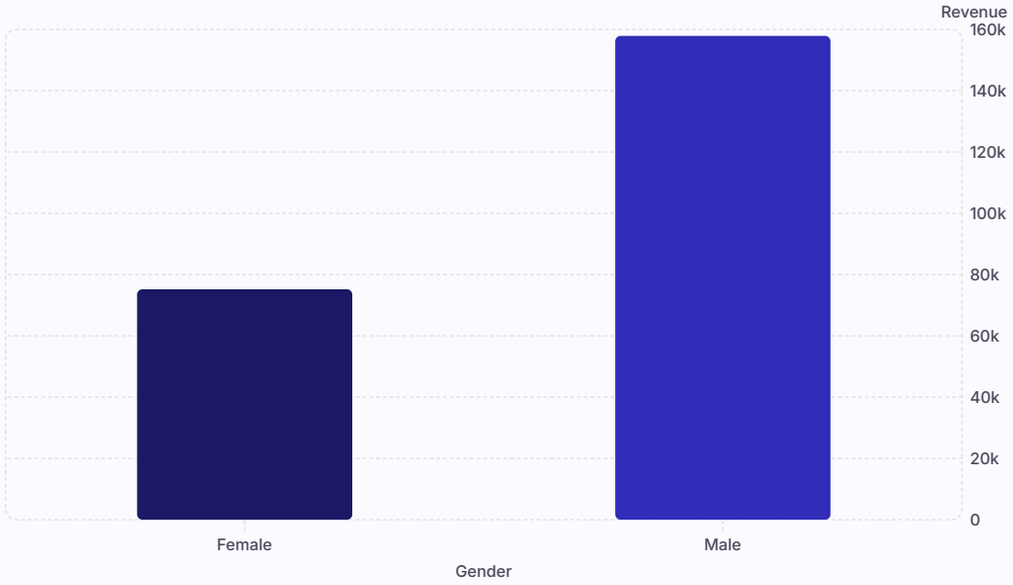
Data Analysis (SQL)

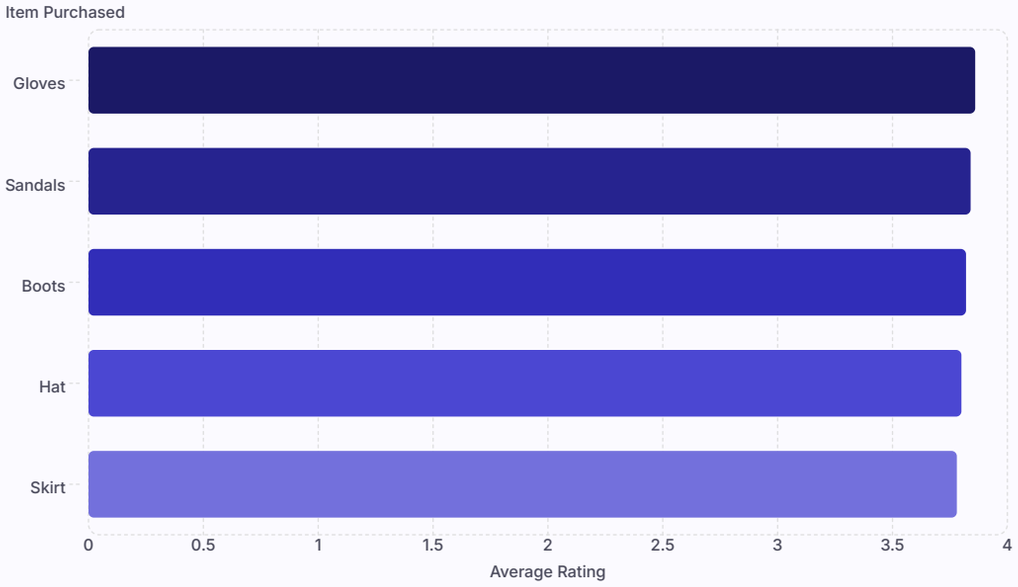
Revenue by Gender

Male customers generated significantly more revenue.



Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, and Skirt received highest average ratings.



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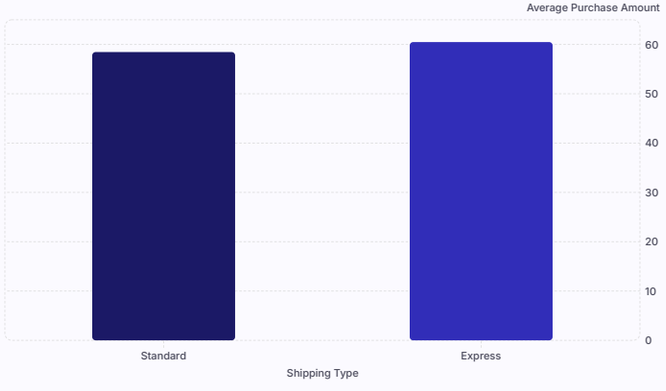
SQL Insights: Spending & Shipping

High-Spending Discount Users

839 customers used discounts but spent above average.

Shipping Type Comparison

Express shipping users have slightly higher average purchase amounts.



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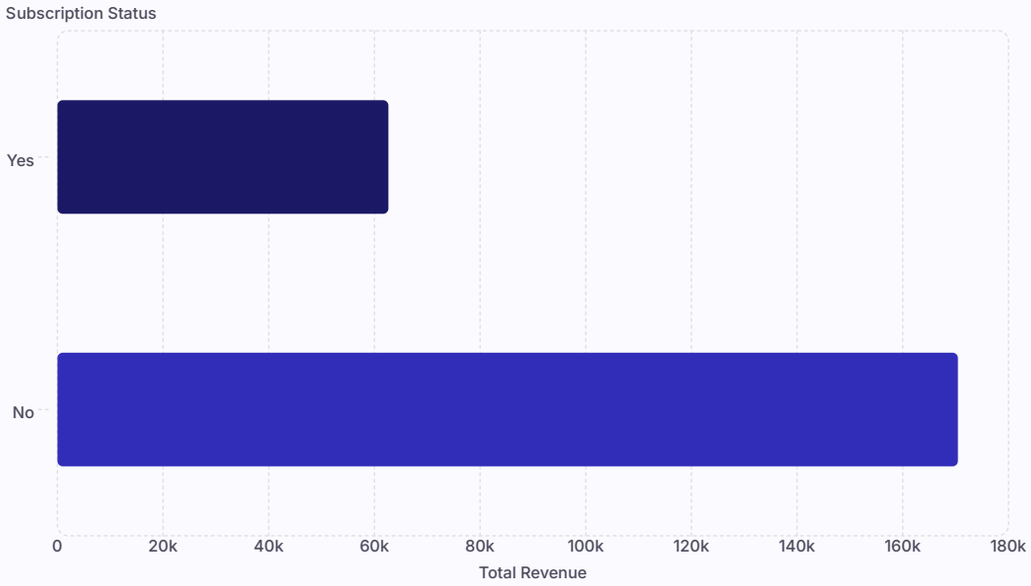
SQL Insights: Subscriptions & Discounts

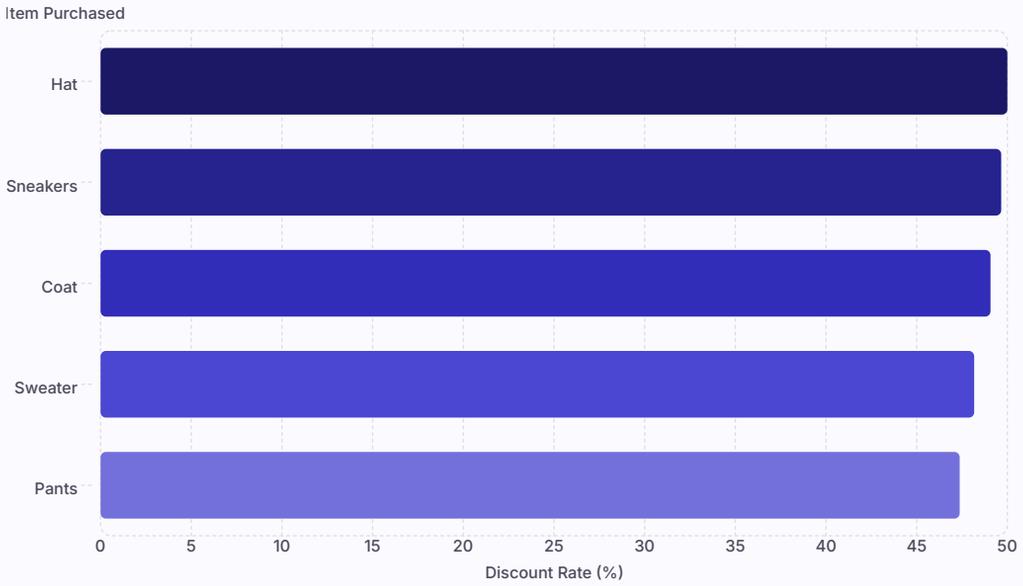
Subscribers vs. Non-Subscribers

Non-subscribers contribute more total revenue, but average spend is similar.



Discount-Dependent Products

Hat, Sneakers, Coat, Sweater, Pants have highest percentage of discounted purchases.



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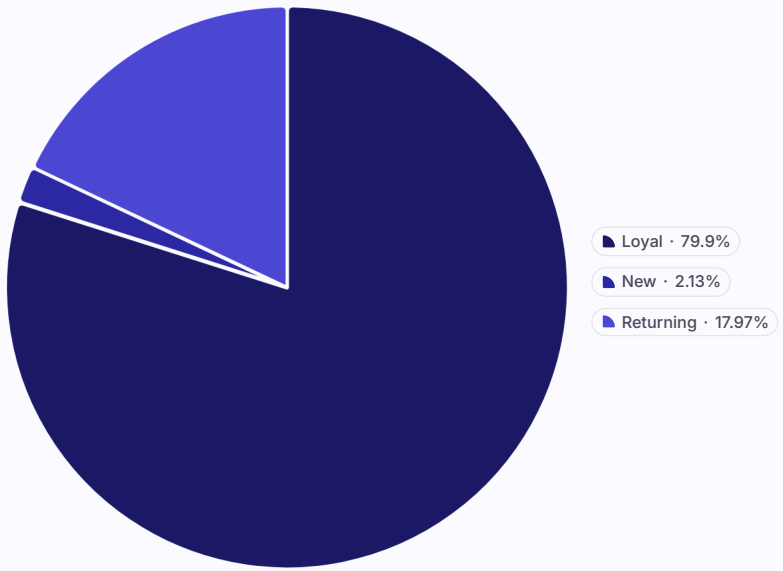
SQL Insights: Segmentation & Age

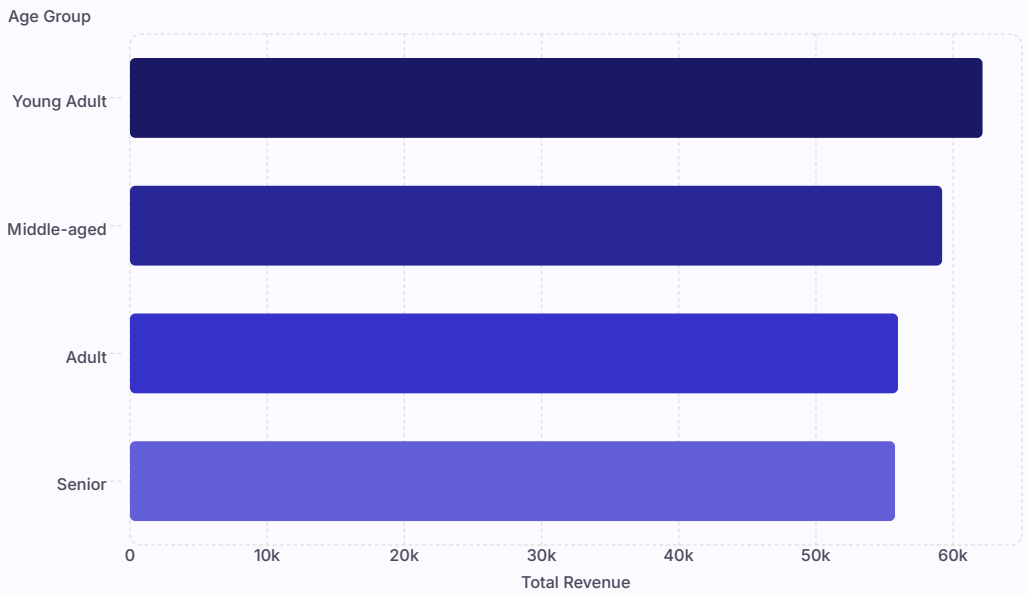
Customer Segmentation

Loyal customers form the largest segment.



Revenue by Age Group

Young adults contribute the most revenue.



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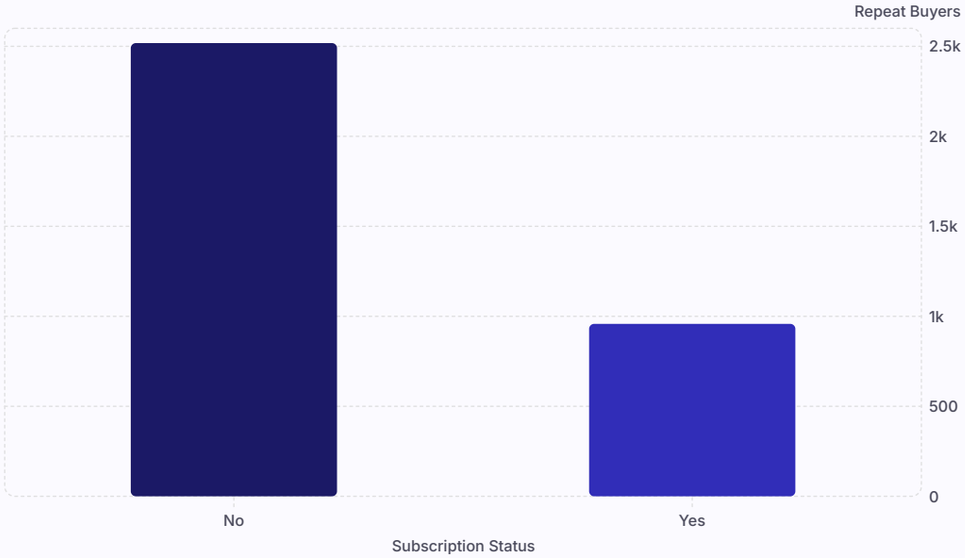
Top Products & Repeat Buyers

Top 3 Products per Category

Identified best-selling items in Accessories, Clothing, Footwear, and Outerwear.

Repeat Buyers & Subscriptions

Repeat buyers are less likely to subscribe.



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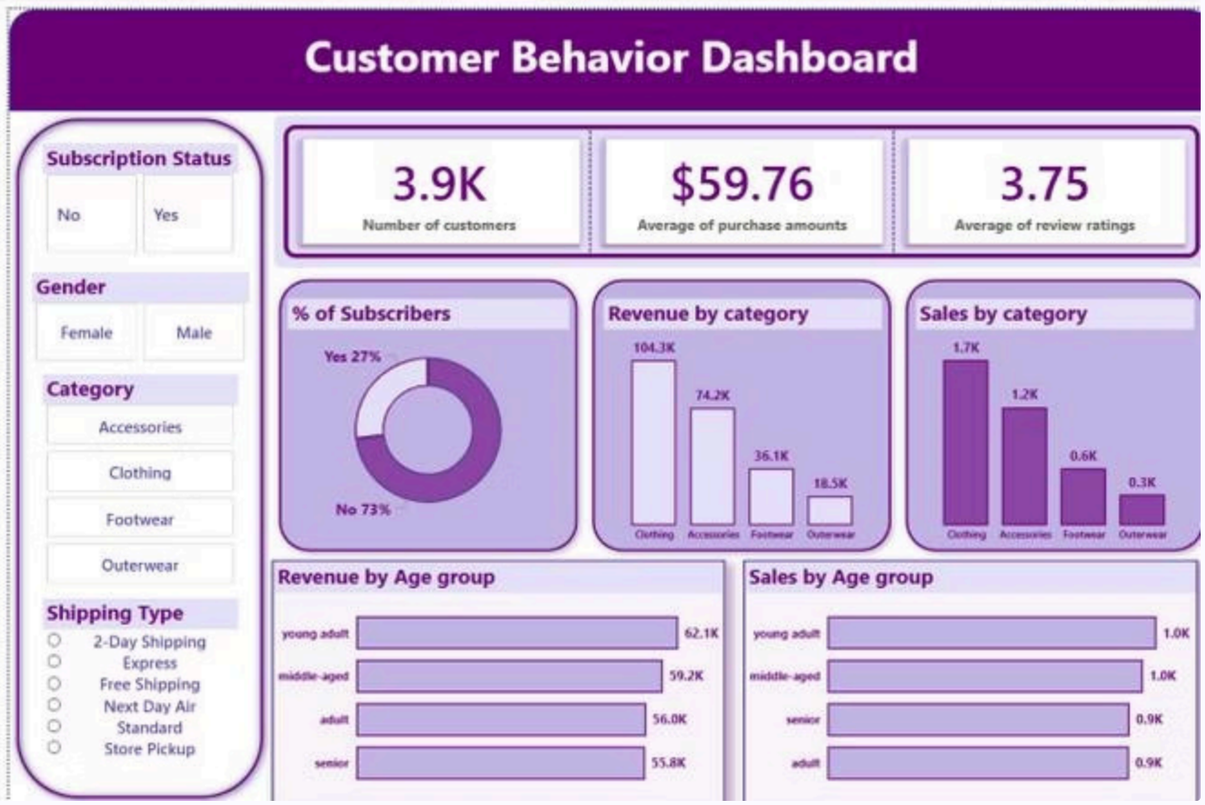
Power BI Dashboard and Business Recommendations



Boost Subscriptions

Promote exclusive benefits.

Customer Loyalty

Reward repeat buyers.

Review Discount Policy

Balance sales with margin.

Product Positioning

Highlight top-rated items.

Targeted Marketing

Focus on high-revenue groups.

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